Air Quality and Income Inequality: Canadian Metropolitan Areas

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Your abstract here.

As climate change reshapes environmental regimes worldwide, academic and political interest in the interactions between climatic and social systems has grown. Environmental exposures, including air pollution, are understood to impact human outcomes (Burke et al., 2022). Studies which evaluate the varied effects of environmental exposures across social and economic subgroups are especially salient (Landrigan et al., 2018). However, the impact of environmental exposures on urban economic conditions is underexplored.

Air pollution is a particularly important source of environmental exposure: globally, "approximately 86% (2.5 billion inhabitants) of urban inhabitants lived in urban areas that exceeded WHO's 2005 guideline annual average $PM_{2.5}$" (Southerland et al., 2022). Previous research has determined that stringent air quality regulations, such as those instated by the Clean Air Act, can exacerbate income inequality (Jha et al., 2019). This conclusion may cause regulatory agencies to avoid implementing environmental policy in order to avoid negative economic outcomes. Under the Clean Air Act, a county that is out of attainment with the American National Ambient Air Quality Standards is subject to "stricter environmental regulations than attainment counties" (Jha et al., 2019).

This detail leads to a potential missattribution of causality in the relationship between strict air quality standards and income inequality. If urban areas with poor air quality are less resilient to increases in income inequality, it could be poor air quality—not air quality regulation—which corresponds with higher relative income inequality. In principle, this destabilizing effect is possible. Poor air quality has a disproportionately negative impact on low-income households (Deryugina et al., 2021). Additionally, poor air quality is associated with worsened labor productivity due to adverse health impacts (Office of Air and Radiation, 2011). Together, these impacts on the workforce might worsen the short- and long-term impacts of an economic shock on income inequality.

A first step to determining whether air quality has a destabilizing effect on income equality is to determine whether a relationship between air quality and income inequality exists. As such, this paper asks if air quality has a statistically significant effect on income inequality within Canadian Metropolitan Areas (CMAs). By illustrating this relationship, we hope to contribute to a robust understanding of the connection between environmental exposures and human outcomes. Assessing the complex interactions between climatic and social systems allows for the implementation of more accurate, equitable, and effective environmental policy. !ADD OVERVIEW OF KEY FINDINGS

I. Data Description

All of the data used in this research was collected from one of three sources: Environment and Climate Change Canada, Statistics Canada's 2021 Census, and Environics Analytics' 2024 Demographic Estimates & Projections.

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Air Quality Health Index (AQHI) is a metric of air pollution created by Environment and Climate Change Canada. Using a risk-weighted scale of ground-level ozone (O_3), particulate matter ($PM_{2.5}$ & PM_{10}) and nitrogen dioxide (NO_2), the index provides a value typically ranging between 1 and 10 (Government of Canada, 2021). AQHI values between 1 and 3 are associated with low health risk; values between 4 and 6 are associated with a moderate health risk; values between 7 and 10 are associated with a high health risk; and values above 10 indicate a very high health risk (Government of Canada, 2015). The AQHI index is typically reported to the public as integer values—but decimal hourly observation data is available from December 2023 until April 2025. Hourly data is organized into CSV files by month of observation and the province of the measurement station. Due to provincial policy, no AQHI data from Quebec is publicly available. We collected all AQHI observation from 2024. We combined each measurement station's hourly averages into a daily average. We also collected a GeoJSON file which contains spatial data about every AQHI measurement station (Environment and Climate Change Canada, 2025). This GeoJSON file was converted to a shapefile.

To associate this data with a given CMA, we collected provincial and CMA boundary shapefiles from Statistics Canada's 2021 Census data (Statistics Canada, 2021a; Statistics Canada, 2021b). While this boundary data is from 2021, CMA boundaries have not changed since the 2021 census. In addition to providing shapefile data, these boundary sets include a census population designation and land area calculation (km^2) for each CMA. Once we adjusted the Coordinate Reference Systems of all relevant shapefiles, we were able to geographically group AQHI measurement stations with CMAs. Not all AQHI measurement stations are located within a CMA, and not all CMAs contain an AQHI measurement station—these incomplete measurements are excluded from our data. Additionally, when a CMA contained multiple AQHI measurement stations, their daily averages were combined. Having combined and organized our daily average AQHI data by CMA, we then calculated each CMA's annual average AQHI and percentage of daily average AQHI values greater than 3. Days without any AQHI reading were omitted from the calculation of these averages. One CMA, Brandon, was ommitted because it had less than 100 days of collected AQHI data, introducing potential seasonal bias in AQHI measurement. We have chosen average AQHI as our independent variable of interest.

Demographic data about every CMA was collected from Environics Analytics' 2024 Demographic Estimates & Projections (Environics Analytics, 2024). Official census data for 2024 does not exist, but this dataset—which was created "in conjunction with experts at The Centre for Spatial Economics"—is one of the best available projections of Canadian demographic data (Environics Analytics, 2024). It was derived from Statistics Canada's 2021 Census data and made available through the SimplyAnalytics platform. We chose to collect the following 2024 data about every CMA:

- Mean Income ($)
- Median Household Income ($)
- Percent of Households with Household Income $100,000 or Over (%)
- Percent of Households with a Household Income of $100,000 or Over with a Household Income of $200,000 or Over (%)
- Percent of Households with a Household Income of $200,000 or Over with a Household Income of $300,000 or Over (%)
- Percent Male Population (%)
- Percent of Population with Visible Minority Status (%)
- Percent of Population 15 Years or Over Married Or Living With A Common-Law Partner (%)
- Percent of Population without Knowledge of English or French (%)
- Percent Immigrant Population (%)
- Percent of Population 15 Years or Over in the Labour Force (%)

- Employment Rate (%)
- Percent of Population 15 Years or Over with a College, CEGEP or Other Non-University Certificate or Diploma (%)
- Percent of Population 15 Years or Over with a University Degree (%)

The CMA demographic data does not include a measurement of income inequality. The Gini Coefficient is a measure of income inequality $G \in [0, 1]$, with $G = 0$ representing perfect income equality and $G = 1$ representing maximum income inequality. We have chosen the Gini Coefficient as our dependent variable. We estimated the Gini Coefficient for each CMA population using the following vector I of 2024 demographic data:

- I_0: Median Household Income
- I_1: Percent of Households with Household Income \$100,000 or Over
- I_2: Percent of Households with a Household Income of \$100,000 or Over with a Household Income of \$200,000 or Over
- I_3: Percent of Households with a Household Income of \$200,000 or Over with a Household Income of \$300,000 or Over

From I we can approximate the Lorenz Curve and corresponding Gini Coefficient of every CMA. We used a monotonically increasing cubic spline interpolation to generate a polynomial representing the income distribution of every CMA. The complete approximation process is detailed in Appendix A.

Two CMAs were excluded from the dataset because of their Gini Coefficients: Wood Buffalo and Yellowknife. Both CMAs have very high mean incomes, which means that our upper bound of household income (\$300,000) is not sufficient to accurately estimate the Gini Coefficient.

No noticable outliers are present in the remainder of our CMA demographic data. All other listed variables acted as control variables, and their purpose will be discussed further in our Model Specification section.

II. Summary Statistics

A map of all relevant AQHI measurement stations and CMAs is available in Appendix B.

Table 1 provides descriptive statistics for the primary variables employed in our empirical analysis. The table summarizes data from CMAs across Canada, including measures of income inequality, air quality, and demographic characteristics.

The mean Gini coefficient is 0.383 with a standard deviation of 0.100, reflecting variation in income inequality across CMAs. On average, the daily Air Quality Health Index (AQHI) is 2.054, with high AQHI days (those exceeding a value of 3) constituting approximately 19.44% of the year. Demographic and socioeconomic characteristics include the percentage of the population that is male, a visible minority, married or in a common-law relationship, employed, and the average income level. Including this data as as control variables in our regression will mitigate confounding factors and better isolate the relationship between air quality and income inequality.

It is helpful to understand the simple correlation between each of these variables. Table 2 displays the correlation between all considered variables.

We began by estimating a simple linear regression of income inequality (Gini Coefficient) on average AQHI, as displayed in Figure 1.

Table 1—: Census Metropolitan Areas: Key Variables ($n = 63$)

Variable	**Mean**	**SD**	**Min**	**Max**
Gini Coefficient	0.383	0.100	0.142	0.586
Average AQHI	2.054	0.287	1.455	2.689
High AQHI Days (%)	19.444	21.932	0.000	85.000
Average Income ($)	56,564.91	7,484.71	43,216.52	77,301.47
Male Population (%)	49.713	0.854	48.156	52.114
Visible Minority (%)	15.394	12.093	2.208	58.503
Married or Common-law (%)	56.331	2.444	48.760	61.884
No English or French (%)	0.902	1.091	0.074	5.639
Immigrant (%)	15.888	9.435	3.017	49.748
Labour Force (%)	51.357	4.267	42.844	62.069
Employment Rate (%)	91.359	2.212	82.300	96.600
College or Diploma (%)	20.529	3.822	14.092	28.680
University Degree (%)	23.312	7.478	11.039	40.393

Note: !ADD DESCRIPTION

Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

Table 2—: Census Metropolitan Areas: Variable Correlation ($n = 63$)

Variable	**G.C.**	**A.A.**	**H.A.D.**	**A.I.**	**M.P.**	**V.M.**	**M.C.**	**N.E.F.**	**I.**	**L.F.**	**E.R.**	**C.D.**	**U.D.**
Gini Coefficient	—	-0.224	-0.267	-0.879	-0.551	-0.475	-0.277	-0.302	-0.527	-0.726	-0.397	0.467	-0.444
Average AQHI		—	0.907	0.377	0.237	0.359	0.019	0.212	0.361	0.230	-0.023	-0.109	0.069
High AQHI Days (%)			—	0.407	0.288	0.393	0.085	0.233	0.431	0.244	-0.009	-0.221	0.110
Average Income ($)				—	0.471	0.360	0.329	0.114	0.405	0.649	0.363	-0.446	0.370
Male Population (%)					—	0.192	0.240	0.178	0.185	0.461	0.184	-0.379	-0.109
Visible Minority (%)						—	0.042	0.880	0.971	0.550	0.135	-0.474	0.668
Married or Common-law (%)							—	0.006	0.083	0.362	0.324	-0.252	0.001
No English or French (%)								—	0.835	0.354	0.089	-0.416	0.472
Immigrant (%)									—	0.569	0.232	-0.506	0.702
Labour Force (%)										—	0.420	-0.349	0.628
Employment Rate (%)											—	-0.286	0.255
College or Diploma (%)												—	-0.283
University Degree (%)													—

Note: !ADD DESCRIPTION

Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

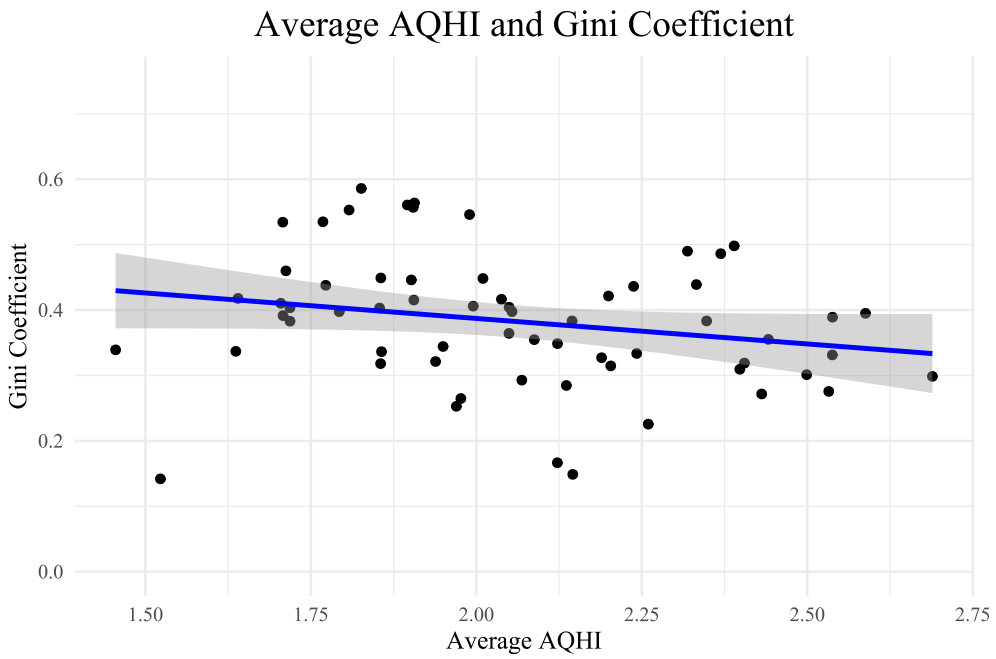


Figure 1. : Scatterplot and Bivariate Regression of CMA Average AQHI and Gini Coefficient

Note: !ADD DESCRIPTION

Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

III. Model Specification

A. *Main Model Specification*

The core objective of this model is to estimate the effect of air quality (as captured by AQHI) on income inequality (as measured by the Gini Coefficient) within CMAs. To investigate the relationship between air quality and income inequality, we employ an ordinary least squares (OLS) regression model:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{(k=1)}^{7} \beta_{k+2} W_{k,i} + \epsilon_i \tag{1}$$

The dependent variable in this model is $Gini_i$, our estimation of the Gini Coefficient. The independent variable of interest is $AQHI_i$, which is the average daily AQHI in a given CMA. This captures typical air quality conditions, accounting for various types of pollutants (Government of Canada, 2021). This model will directly test whether *ceteris paribus* differences in air quality are associated with changes in the Gini Coefficient. For example, if the estimated coefficient β_1 on $AQHI_i$ is significant and positive, that would indicate that worse air quality is correlated with higher income inequality within a CMA. We also have included an interaction variable constructed from $AQHI_i \times IncomeCat_i$, which is a categorical variable representing the mean income for a given CMA. We include this variable because we expect air quality to effect income inequality differently in places with higher or lower average incomes due to the heterogenous health impacts of air pollution (Deryugina et. al, 2021). Adding in this interaction variable will not introduce

endogeneity: mean income was not used in our calculation of the Gini coefficient, and the interaction variable only accounts for changes in the effect of $AQHI_i$ on $Gini$ across different levels of average CMA income. Since we are not including average income directly, we have centered and categorized CMA mean incomes to make our interpretation of the model more clear and avoid any problems caused by potential non-linearity in the interaction variable.

Control variables are categorized into three main groups: socio-demographic, economic, and educational control variables. Our chosen socio-demographic variables are the percentage of males, the percentage of visible minorities, and the percentage of immigrants. There is higher male representation in industries like manufacturing, which are correlated with both air pollution and income inequality (Varechova et al., 2016). There are also higher concentrations of visible minorities and recent immigrants in low-income, high-pollution areas where income inequality is often the greatest (American Lung Association, 2023). We also include marital status: in lower-income regions, dual-income households often exhibit higher financial stability than single-income households (Pais et al., 2014).

Our chosen economic control variables are employment rate and labor force participation rate, which adjust for income polarization.

Educational control variables are included to address barriers to economic participation. We chose to include both the percentage of a CMA's population aged 15 years or older who hold a University degree and the percentage of a CMA's population aged 15 years or older who hold a College, CEGEP or other non-University certificate or diploma. We also chose to include the percentage of individuals in a CMA who do not speak English or French. While we considered all of these variables as candidate control variables, we ultimately decided to omit two to account for high multicollinearity. The percentage of immigrants in total population has an extremely high correlation with both the percent of population with visible minority status ($\rho = 0.971$) and the percent of population without knowledge of English or French ($\rho = 0.835$). As such, both of these control variables were not included in our regression.

The error term represents unobserved factors influencing local income inequality that are not explicitly included in the model.

B. Model Variations

To ensure the validity of our results, we propose five alternative models.

We first add $AQHI_i^2$ to our regression to account for a possible nonlinear effect of air quality on income inequality. Different levels of environmental exposure to poor air quality might be associated with varying health and economic consequences: by including both a linear and a quadratic term, we can test whether the effects of higher pollutant concentrations follow a simple linear pattern or exhibit diminishing/increasing marginal effects (Banzhaf & Walsh, 2008; Landrigan et. al, 2018). This gives us the following alternative model:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i^2 + \beta_3 AQHI_i \times IncomeCat_i + \sum_{(k=1)}^{7} \beta_{k+3} W_{k,i} + \epsilon_i \tag{2}$$

Next, we replace the interaction variable $AQHI_i \times IncomeCat_i$ with the quadratic term $AQHI_i^2$. This change tests whether AQHI's effect on income inequality actually varies in CMAs with higher or lower average incomes: it's possible that a direct nonlinear effect of air quality on inequality is

obscured by multicollinearity with the interaction variable. This gives us an alternative model:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i^2 + \sum_{(k=1)}^{7} \beta_{k+2} W_{k,i} + \epsilon_i \tag{3}$$

Next, we remove the control variables for university degree attainment and college, CEGEP or other non-university certificate or diploma attainment to verify that our results are not unduly influenced by the inclusion of education measures and that we are not overfitting our regression. This gives us the following alternative model:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{(k=1)}^{5} \beta_{k+2} W_{k,i} + \epsilon_i \tag{4}$$

Finally, we incorporate CMA fixed effects—census population designation and geographic size—to capture potential spatial heterogeneity. This approach address concerns that more populous or geographically large CMAs might differ in both pollution levels and income inequality. This gives us the two models of the form

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{(k=1)}^{8} \beta_{k+2} W_{k,i} + \epsilon_i \tag{5}$$

which include a control variable for CMA designation and land area respectively.

IV. Table of Results

!NEEDS UPDATE

V. Discussion

References here (manual or bibTeX). If you are using bibTeX, add your bib file name in place of BibFile in the bibliography command.

A: Mathematical Analysis

β_1

Table 3—: Main vs. Robustness Checks on Determinants of Income Inequality Across Canadian CMAs ($n = 63$)

	Dependent Variable: Gini Coefficient					
	Main	Alt 1	Alt 2	Alt 3	Alt 4	Alt 5
Equation Number	(1)	(2)	(3)	(4)	(5)a	(5)b
Intercept	2.964***	3.637***	3.691***	2.766***	2.942***	2.940***
	(0.557)	(0.671)	(0.701)	(0.343)	(0.604)	(0.557)
Average AQHI	0.082*	−0.478	−0.491	0.085**	0.081*	0.081**
	(0.031)	(0.269)	(0.357)	(0.028)	(0.031)	(0.028)
(Average AQHI)^2	—	0.137*	0.117	—	—	—
		(0.064)	(0.085)			
Male (%)	−0.046***	−0.048***	−0.039**	−0.042***	−0.46***	−0.049***
	(0.012)	(0.012)	(0.012)	(0.008)	(0.012)	(0.011)
Immigrant (%)	−0.002	−0.002	−0.001	−0.002*	−0.002	−0.001
	(0.001)	(0.001)	(0.003)	(0.001)	(0.001)	(0.001)
Married (%)	0.003	0.001	−0.001	0.003	0.003	0.002
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Employment Rate (%)	−0.003	−0.03	−0.004	−0.003	−0.003	−0.02
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
University (%)	−0.001	−0.001	−0.002	—	−0.001	−0.001
	(0.002)	(0.002)	(0.002)		(0.002)	(0.002)
College or Diploma (%)	−0.001	0.000	0.003	—	−0.001	0.001
	(0.002)	(0.002)	(0.003)		(0.002)	(0.002)
Labour Force (%)	−0.003	−0.003	−0.008*	-0.004	−0.004	−0.003
	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)
Average AQHI: Medium Income	−0.032***	−0.030***	—	−0.032***	−0.033***	−0.036***
	(0.008)	(0.008)		(0.008)	(0.009)	(0.009)
Average AQHI: High Income	−0.060***	−0.061***	—	−0.060***	−0.062***	−0.062***
	(0.008)	(0.008)		(0.008)	(0.009)	(0.008)
Region Fixed Effects	—	—	—	—	Designation	Land Area
R^2	0.794	0.806	0.647	0.793	0.795	0.804
Adjusted R^2	0.754	0.764	0.587	0.762	0.746	0.762

Note: Robust standard errors are listed in parentheses.
Coefficient estimate significance is indicated as follows: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
This table presents the results of OLS regressions analyzing the relationship between air quality (as measured by AQHI) and income inequality (Gini coefficient) across Canadian Census Metropolitan Areas (CMAs). The main model (Column 1) uses the average AQHI as the key independent variable, while alternative specifications (Columns 2–4) test robustness by replacing AQHI with the percent of days with high AQHI, dropping major outliers (Toronto and Vancouver), and including region fixed effects, respectively. Across all models, the share of male population and the percentage of visible minorities are significantly and negatively associated with inequality. The adjusted R-squared improves considerably with the inclusion of regional controls, suggesting that location-based unobservables play a substantial role in explaining income inequality.

Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.